

VIJAY PRAJWAL RAJASEKAR

Mechanical Engineering

AEROSPACE / MECHANICAL DESIGN FOCUS

Mechanical engineering student interested in aerospace design, with hands-on experience in CAD, composite manufacturing, and engineering analysis.

EMAIL

vj0501@student.ubc.ca

LINKEDIN

[linkedin.com/in/vijayprajwalrajasekar/](https://www.linkedin.com/in/vijayprajwalrajasekar/)

LIVE PORTFOLIO

<https://vijayprajwal.github.io/VijayPrajwalRajasekar-Portfolio/>



SCAN TO VIEW INTERACTIVE LIVE PORTFOLIO & 3D CAD MODELS

Forged Carbon Fiber Motor Mount

ROLE: Mold design & layup lead TOOLS: SolidWorks, hand layup



FINISHED CARBON FIBER MOTOR MOUNT (TOP-DOWN VIEW)

57.9%

MASS REDUCTION

114 g (Aluminum) → 48 g
(Carbon)

OBJECTIVE

Replace 114 g aluminum mounts with 48 g carbon fiber parts — a 57.9% mass reduction — without changing the motor bolt pattern.

DESIGN & APPROACH

Modeled a drop-in replacement in SolidWorks preserving the original motor bolt pattern, then designed a two-part aluminum mold. Hand-packed chopped carbon tow and resin into the mold, refining the layup through multiple iterations.

ITERATIVE MANUFACTURING

Resolved mold issues through testing: added draft angle to fix parts sticking, repositioned vents to address resin starvation, and adjusted clamping to reduce uneven flash.

OUTCOME

Finished mount weighed 48 g — a 57.9% reduction from the original 114 g aluminum part.



48 G MASS VALIDATION ON SCALE

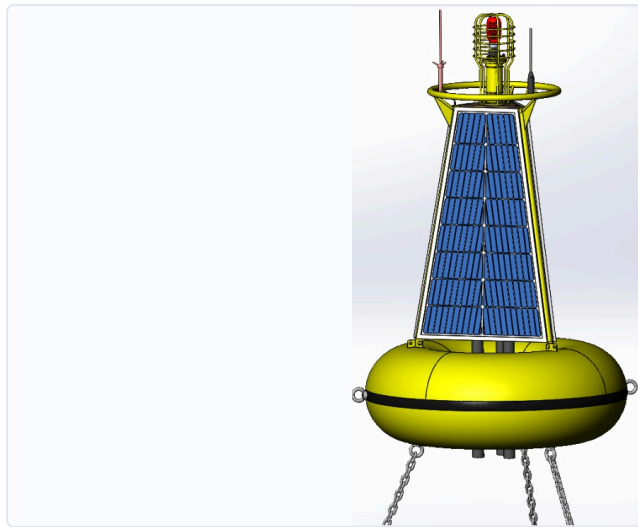


TWO-PART MOLD HALVES OPEN

Data Acquisition Buoy System (DABS)

ROLE: CAD Design & Engineering Analysis

TOOLS: SolidWorks



DABS BUOY SYSTEM ASSEMBLY CAD RENDER

NOMINEE

DESIGN PROCESS CATEGORY
Aquahacking Binational 2026

OBJECTIVE

Design a buoy system to monitor nitrogen and orthophosphate levels in aquatic environments.

CAD & DESIGN APPROACH

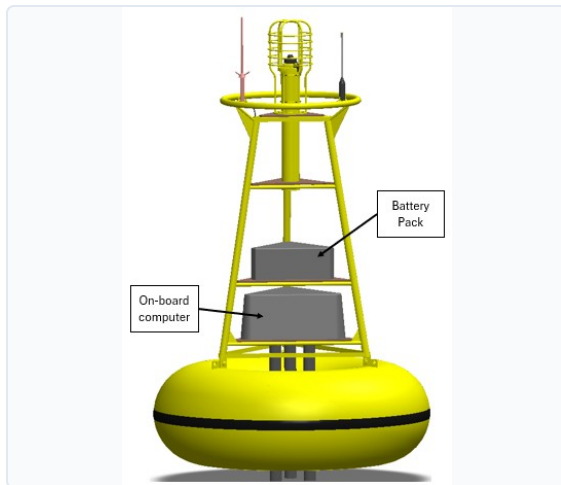
Iterated the SolidWorks CAD to improve component layout and fit the electronics and sensors inside the housing. Exported a 3D browser viewer for stakeholder review.

ENGINEERING ANALYSIS

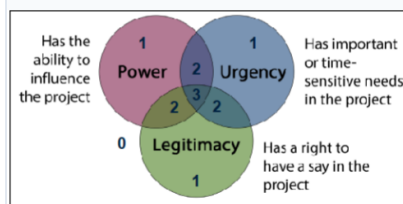
Conducted a Stakeholder Salience Model (Power, Legitimacy, Urgency) and a Risk Assessment Matrix covering risks such as biofouling and watertight seal degradation.

OUTCOME

Project nominated for the Design Process Category at Aquahacking Binational 2026.



INTERNAL COMPONENT LAYOUT CAD



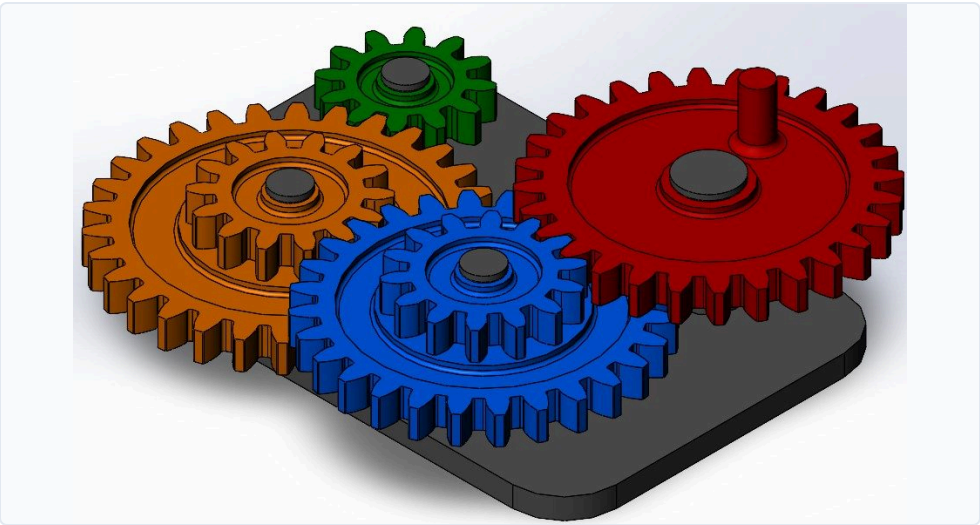
STAKEHOLDER SALIENCE MODEL

Data Acquisition Buoy System (DABS) - Prototype 1									
		Degree of Urgency		Degree of Legitimacy		Degree of Power		Overall Risk	
Stakeholder	Category	Urgency	Power	Legitimacy	Power	Urgency	Legitimacy	Significance	Severity
Local Government	Regulatory	High	High	High	High	High	High	High	High
Local Community	Stakeholder	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium
Local Industry	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low
Local Academia	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low
Local Media	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low
Local NGOs	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low
Local Businesses	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low
Local Residents	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low
Local Tourists	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low
Local Researchers	Stakeholder	Low	Low	Low	Low	Low	Low	Low	Low

RISK ASSESSMENT MATRIX

Involute Gear Train

ROLE: Personal build TOOLS: SolidWorks (equation-driven), Prusa i3 MK3S, PLA



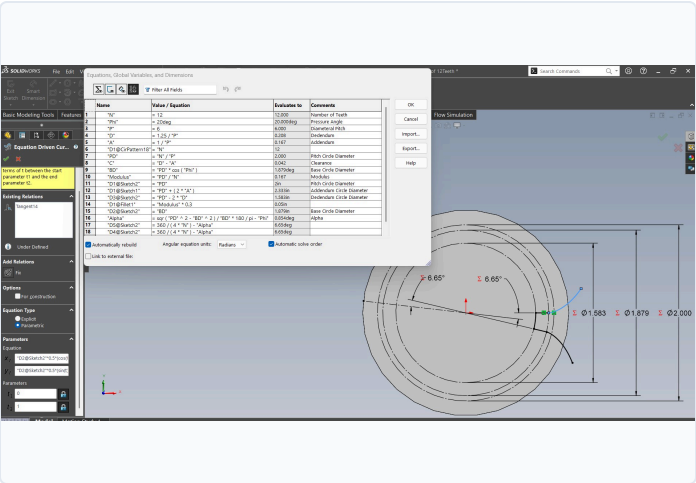
INVOLUTE GEAR TRAIN ASSEMBLY ISOMETRIC CAD

OBJECTIVE

Design and build a physical gear train with a 10:1 overall ratio across three reduction stages.

EQUATION-DRIVEN DESIGN

Built the involute tooth profile in SolidWorks using Equation Driven Curves and Global Variables so the gear geometry could be adjusted parametrically. Each gear ratio stage was defined by equations rather than fixed dimensions.



SOLIDWORKS EQUATIONS MANAGER

MANUFACTURING & CLEARANCE

Printed all gears on a Prusa i3 MK3S in PLA. Tested fit after each print and dialed in 0.15 mm clearances across stages to achieve smooth rotation without binding.

OUTCOME

Achieved a 10:1 overall gear ratio across three stages (12T→30T, 14T→28T, 14T→28T). The assembled train runs smoothly with no binding.

10:1

GEAR RATIO

3 reduction stages (smooth, no binding)



FDM PRINTED GEARS ON PRUSA I3 MK3S